Machine Learning for Vehicle Signal Analytics



Presented by

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VOLVO

THE MOTIVATION

Why Machine Learning?

Every day we collect thousands of vehicle signals.

- Torque
- Motor speed
- Battery voltage
- Temperature
- Brake pressure
- Accelerator pedal
- Steering angle

How can we automatically discover useful information from these signals

?

Machine Learning is one answer.

Manual limit: Time consuming, too slow to scale.
ML : finds patterns automatically, Predication, Early Fault detection.

WHERE WE ARE TODAY

The Current Workflow



What happens when we have...

500
vehicles

50 TB
of measurements

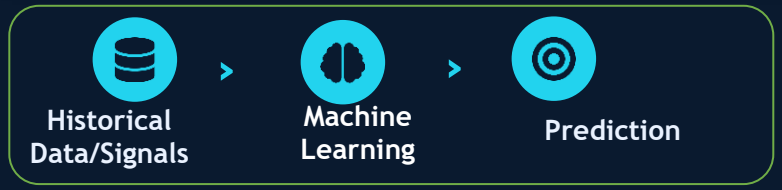
Millions
of signals

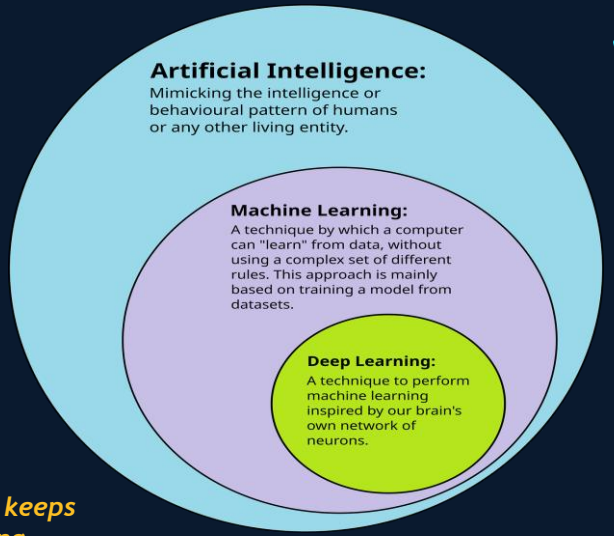
Manual analysis becomes impossible.
ML helps engineers work faster — it doesn't replace them.

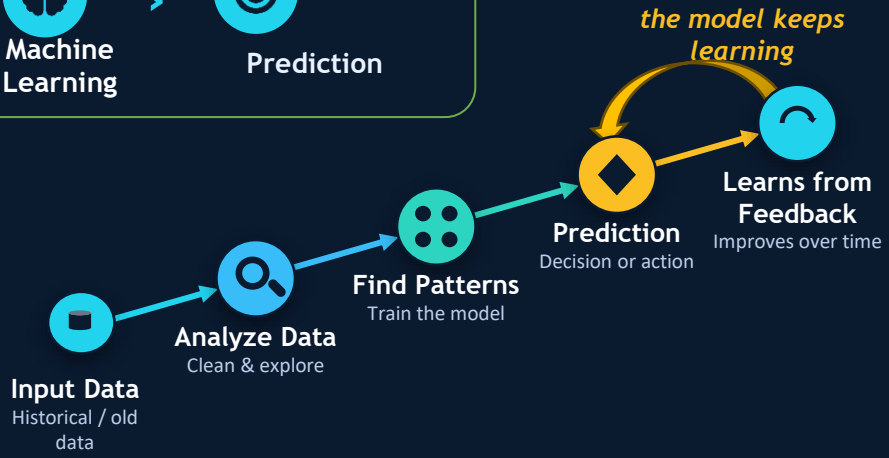
THE CORE IDEA

What is Machine Learning?

Algorithms that learn patterns from historical data to make predictions or decisions — without being explicitly programmed.







THREE FAMILIES

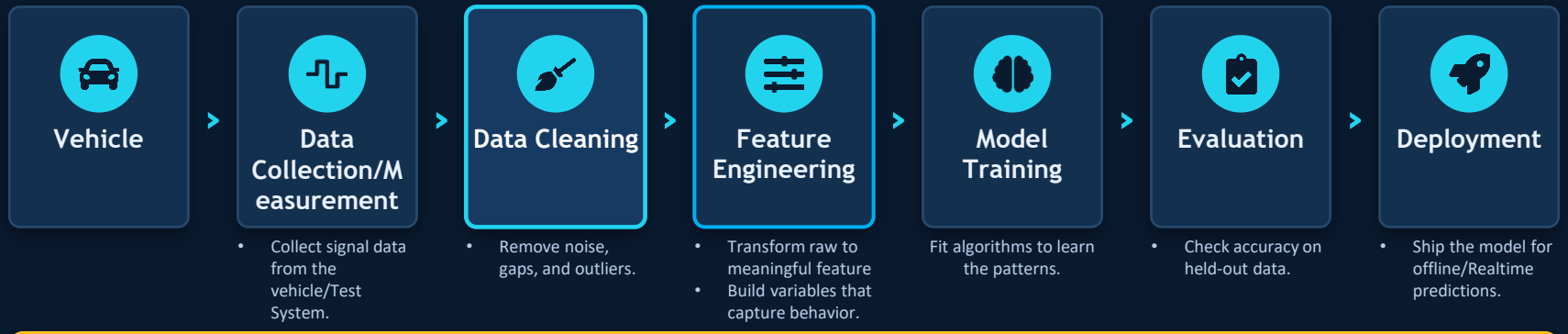
Types of Machine Learning

TYPE	WHAT IT MEANS	EXAMPLE
Supervised 	• Trained on labeled data where the correct answer is already known. • Has Labels	Predict overtorque
Unsupervised 	• Finds hidden patterns and structure in data that has no labels. • No Labels	Detect unusual driving
Reinforcement 	• Learns by trial and error, guided by rewards and penalties.	Autonomous driving

Supervised learning
is the most common approach in vehicle analytics.

END TO END

The Machine Learning Workflow

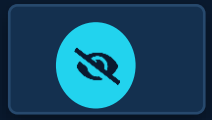


Most projects fail because of poor data quality, not poor algorithms.

Cleaning and feature engineering are where the real work happens.

THE HIDDEN REALITY

Data Cleaning



Missing values



Noise



Synchronization



Duplicate signals

70-80%
of a data scientist's time is spent preparing data

Clean signals in, reliable models out
this is where engineers add the most value.

Feature Engineering

- **Transform** raw data into insight of model
- Instead of raw **Torque**, Calculate :

- Maximum torque
- Mean torque
- Torque ripple
- Standard deviation
- Peak speed
- Acceleration

Features often matter more, A simple model with strong features usually beats a complex model fed raw signals.

THE TOOLBOX

Common ML Algorithms

Linear Regression

Predict Continues Values, Simple

Decision Trees

Transparent rule-based splits

Random Forest

Robust classification/ensemble of trees

XGBoost

High-accuracy gradient boosting/tabular data

Support Vector Machine

Strong decision boundaries

Neural Networks

Complex, non-linear patterns

Different problems require different models.

REAL EXAMPLES

Example: *Over-torque & Anomaly Detection*

Over-torque model

Predict overtorque before it happens

Torque · Speed · Temperature · Pedal

↓

Signal Files → Python → Features → **Random Forest**

↓

Will overtorque occur?

YES **NO**

Anomaly detection

ML flags these automatically:

- Unexpected torque spikes
- Sensor failures
- Abnormal vibration
- Electrical faults
- Unexpected temperature increase

A natural fit: engineers already look for these — ML does it continuously.

DID THE MODEL WORK?

Evaluation

- Evaluate using data the model has never seen before

Accuracy
(TP + TN) / Total
Percentage of correct predictions.,
97

Precision
TP / (TP + FP)
Correct positive calls, how often is it correct !
79.2

Recall
TP / (TP + FN)
Real faults caught, how many faults
95

F1-score
2 · P · R / (P + R)
Balances precision and recall
86.4

Confusion Matrix
TP FP / FN TN
Predicted vs. actual

Actual /120	Predict	Result
over torque	over torque	95
normal	Over torque	25
Over torque	Normal	5

In fault detection, not all errors are equal.

Missing a real fault can be far more costly than raising a false alarm — so recall often matters more than raw accuracy.

PUTTING IT TO WORK

Some Use Cases

- Predict overtorque
- Battery health prediction
- Motor fault detection
- Remaining useful life
- Predictive maintenance
- Energy consumption optimization
- Driver behavior analysis
- Vehicle diagnostics

WHAT MAKES IT HARD

Challenges

- High-frequency signals
- Large datasets
- Noise
- Missing values
- Class imbalance
- Labeling faults
- Model interpretability
- Real-time inference

These are exactly the issues engineers meet every day.

THE PRACTITIONER'S STACK

Tools





Python
Core programming language



NumPy
Numerical array computing



Pandas
Data analysis toolkit



Scikit-learn
Classic ML algorithms



TensorFlow
Deep learning framework



PyTorch
Deep learning framework



Keras
Neural network API



Jupyter Notebook
Interactive notebooks



Git
Version control

IN SUMMARY

Key Takeaways

Machine Learning…

- learns patterns
- scales to massive datasets
- supports engineers
- predicts failures
- automates analysis
- enables smarter vehicles

Questions?

Thank you!

